

Flask Web Server on Alpine Linux (Vultr)

Goal:

Deploy a Flask app using Docker on an Alpine VPS (Virtual Private Server)
and auto-deploy using GitHub

Step 0: Check you have Docker in your System

Step 1: Install Docker on Alpine

```
apk update  
apk add docker docker-cli  
rc-update add docker boot  
service docker start
```

Make sure Docker starts on reboot.

Step 2: Create a Flask App (Locally)

Local project structure:

```
my-flask-app/  
├── app.py  
├── requirements.txt  
└── Dockerfile
```

app.py

```
from flask import Flask
app = Flask(__name__)
@app.route('/')
def hello():
    return "Hello from Flask!"
```

requirements.txt

flask

Step 3: Dockerfile for Flask

Dockerfile

```
FROM python:3.12-alpine
WORKDIR /app
COPY requirements.txt ./
RUN pip install -r requirements.txt
COPY . .
CMD ["flask", "run", "--host=0.0.0.0"]
```

Step 4: Build and Run Docker Container

Make sure you are in the docker

```
docker build -t flask-app .  
docker run -d -p 80:5000 flask-app
```

App will be available on <http://your-vultr-ip/>

Step 5: Push Code to GitHub

1. Initialize git:

```
git init  
git remote add origin <your-repo-url>
```

After the initialization, run the `push_code.sh` script to push the code to github repository.

```
git add .  
git commit -m "Initial commit"  
git push -u origin main
```

Step 6: Auto-Pull + Restart Script

On Alpine server, create this shell script:

```
/home/user/deploy.sh
```

```
#!/bin/sh
cd /home/deploy/my-flask-app
git pull origin main
docker build -t flask-app .
docker stop flask-running
docker rm flask-running
docker run -d --name flask-running -p 80:5000 flask-app
```

Make it executable:

```
chmod +x /home/user/deploy.sh
```

⚙️ Step 7: Auto Git Pull (via GitHub Webhook)

1. On your server:

```
apk add git busybox-suid
```

```
~ # uname -m  
x86_64
```

```
wget https://github.com/adnanh/webhook/releases/latest/download/webhook-linux-amd64.tar.gz  
tar -xzf webhook-linux-amd64.tar.gz  
sudo -n mv webhook /usr/local/bin/  
sudo -n chmod +x /usr/local/bin/webhook
```

2. Install `webhook` (Go-based tool):

```
wget https://github.com/adnanh/webhook/releases/latest/download/webhook-linux-arm64.tar.gz  
tar -xzf webhook-linux-arm64.tar.gz  
sudo -n mv webhook /usr/local/bin/  
sudo -n chmod +x /usr/local/bin/webhook
```

Step 8: GitHub Webhook Settings

1. Go to your GitHub repo → **Settings** → **Webhooks**

2. Add webhook:

- Payload URL: `http://<your-ip>:9000/hooks/flask-hook`
- Content Type: `application/json`
- Secret (optional, but recommended)

Done!

- Local Dev → Push to GitHub
- GitHub Webhook → Alpine pulls + redeploys
- Flask runs inside Docker
- Fully automatic workflow!

Bonus: Enable on Boot

Add to `/etc/local.d/start-webhook.start`

```
#!/bin/sh
cd /home/user/my-flask-app
/usr/local/bin/webhook -hooks hooks.json -port 9000 &
```

Make it executable:

```
chmod +x /etc/local.d/start-webhook.start
rc-update add local
```

Q&A or Next Steps?

- HTTPS with Nginx + certbot
- Use Gunicorn instead of `flask run`
- Add CI/CD with GitHub Actions